

# Separating the known from the unknown at Marfa, Texas: line of sight, refraction, and photometry of the ordinary lights seen from the Marfa lights viewing area

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For more than eighty years, visitors to a roadside viewing area near Marfa, Texas, have reported lights on the horizon that appear, brighten, drift, and vanish. Any attempt to study them faces a foreground of ordinary lights—vehicle headlamps and tail lamps, trains, obstruction lights on towers, and the glow of distant towns—that must be removed before an unusual light can be recognized. We build a quantitative model of that foreground from public data and first principles, and present it as a screening tool. Exact ellipsoidal geometry on 1-m airborne lidar terrain, with atmospheric refraction, gives where each ordinary source can be seen; a single number, the critical refraction coefficient  $k_{\text{crit}} = \max_i 2Ry_i/[s_i(D - s_i)]$ , decides whether a lamp clears the terrain. Because the sight lines graze the ground within about a meter, a Monte Carlo treatment of terrain error, vegetation from the lidar point cloud, and ray tracing through nocturnal inversions are needed to make the answer robust. Market-weighted headlamp beam patterns, federal signal-lamp photometry, and a contrast-threshold model of the eye give how bright each source looks. About 10 km of US Highway 67, 18–40 km away, is in view, always just below the skyline. There a northbound car is typically a magnitude +2.6 light that can outshine Sirius; it appears in 18 separate windows of median 17 s, drifting about  $0.9^\circ$  per minute and brightening and fading as the road’s grade changes. Southbound tail lamps are near the limit of vision. We check the geometry against photographs from the platform, tabulate the observable signatures of every catalogued source, and map the parts of the view where no catalogued source can appear. An online version lets observers test a sighting against the model.

## I. INTRODUCTION

About 14 km east of the town of Marfa, Texas, a roadside viewing area on US Highway 90 looks south and southwest across a high grassland basin toward the Chinati Mountains. Since at least the 1940s, visitors have reported lights there that appear on the horizon, brighten, move, split, and vanish.<sup>1,2</sup> Physicists have studied them before. Stephan *et al.* combined azimuth logging, video, and continuous spectroscopy over 20 nights and used them to identify vehicle headlamps, distant streetlamps, and fires among the lights recorded; they also showed that atmospheric oxygen absorption can measure the distance to such sources.<sup>3</sup> Similar “ghost lights” elsewhere have been traced to headlights and locomotives (Brown Mountain, North Carolina;<sup>4</sup> the Paulding light, Michigan<sup>5</sup>) or to a car seen by mirage (the Min Min light, Australia<sup>6</sup>), whereas at Hessdalen, Norway, a long instrumental campaign left some events unexplained.<sup>7</sup>

Any search for a rare phenomenon against a bright background must first remove the background. Astronomers searching for new transients subtract catalogued stars and mask known artifacts before they look at what remains. In the language of detection theory, ordinary sources that are mistaken for the phenomenon are false positives, and what survives the screening is the

list of candidates. At Marfa the background is made of ordinary lights: headlamps and tail lamps on the highways, locomotive headlights, the red obstruction lights of communication towers, the glow of distant towns, and aircraft. Without a quantitative model of where and how these appear, a report of a light can be neither dismissed nor taken seriously, and observers with different expectations talk past each other.

This paper builds that model, and we intend it as a tool rather than as an argument. For every catalogued ordinary source it predicts where in the view the source can appear, how bright it should be, what color it is, and how it should behave in time. An observer can then remove the lights that match an ordinary source and concentrate on those that do not. We do not claim that all Marfa lights are ordinary; a model of the known cannot say what the unknown is. What it can do is define the observables that would make a light interesting: a position where no catalogued source can appear, or a behavior that no catalogued source shows.

The model is built from first principles and public data, and it is deliberately a model paper: we report no new field experiments. We answer three questions. (i) *Geometry*: which parts of which roads, railroads, and structures are in direct view of the platform, allowing for Earth’s curvature, the terrain, vegetation, and refraction? (ii) *Photometry*: how bright is a car there, given where the platform falls in its beam? (iii) *Behavior in time*: what does one car look like as it drives the road? Each step uses physics taught in undergraduate

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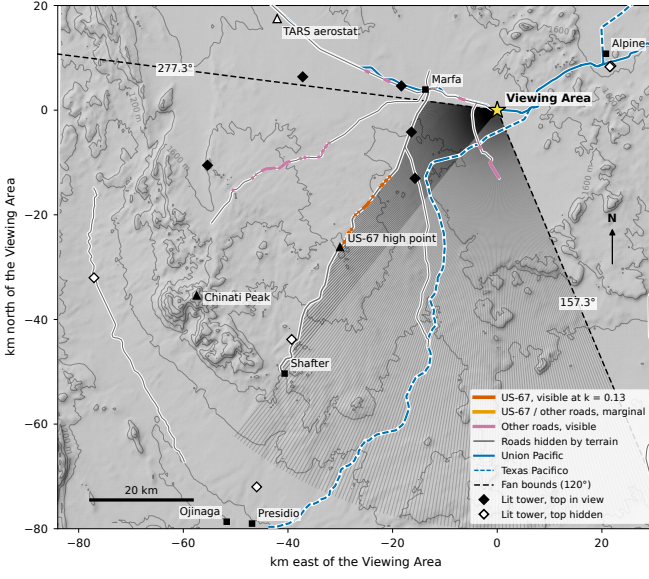


FIG. 1. The view from the Marfa lights viewing area (star) toward the southwest. Roads are colored where a headlamp is in direct view of the platform (Sec. VI). Diamonds are lit communication towers. The shaded fan shows rays every  $0.5^\circ$  between  $157.3^\circ$  and  $277.3^\circ$  true.

courses (geodesy and curvature, refraction in a stratified medium, the inverse-square law and the astronomical magnitude scale, and the statistics of visual detection), and each rests on data a student can download. We give simple estimates alongside the full calculation.

Section II describes the site and data, Sec. III the line-of-sight model and its robustness, Sec. IV refraction beyond a single coefficient, and Sec. V the photometry. Section VI gives the results, including a check against photographs, and Sec. VII turns them into a screening procedure. Section VIII discusses what the model does and does not show. Full methods are in the Supporting Information.<sup>8</sup>

## II. SITE AND DATA

The eye is placed 1.6 m above the viewing platform at  $30.27511^\circ\text{N}$ ,  $103.88280^\circ\text{W}$ . The lidar resolves the platform as a pad 1.3–1.5 m above the surrounding grassland, at 1494.80 m above the North American Vertical Datum (NAVD88). Figure 1 shows the region and the  $120^\circ$  fan of bearings usually watched.

*Terrain.* The U.S. Geological Survey’s 1-m digital elevation model from the lidar project TX\_WestTexas.2018.D19 (27 tiles of  $10 \times 10$  km, collected between February and May 2019) covers every sight line of interest.<sup>9</sup> Its check-point test gives a vertical root-mean-square error of 5.6 cm on open ground (565 points) and a 95th-percentile error of 21.6 cm in vegetation (461 points). Outside the lidar we use the  $1/3$  and 1 arc-second products. Heights are converted to

ellipsoidal heights with the GEOID12B model, the one the lidar was processed with (GEOID18 differs by at most 6 cm here).<sup>10</sup> From the lidar point cloud we also build a map of shrubs, fences, and buildings along the grazing corridors.

*Roads and traffic.* Road centerlines come from the Texas Department of Transportation (all 821 public-road features in the area), and traffic from its annual average daily traffic (AADT) counts.<sup>11</sup> On US-67 between Marfa and the village of Shafter the 2025 AADT is 1121 and 1383 vehicles per day at the two count stations.

*Lamps, air, and eyes.* Beam patterns, signal-lamp photometry, atmospheric clarity, and the detection threshold are described in Sec. V.

## III. LINE OF SIGHT

### A. An estimate

For a smooth sphere of radius  $R$ , a point at distance  $d$  lies below the observer’s horizontal plane by  $d^2/2R$ . Refraction bends near-horizontal rays downward, concave toward the ground, with curvature  $k/R$ ; the conventional “standard” value of the refraction coefficient is  $k \approx 0.13$ .<sup>12</sup> The effective drop is therefore

$$\Delta h = \frac{(1 - k) d^2}{2R}. \quad (1)$$

At  $d = 30$  km this is 61 m, and the horizon of a smooth Earth for an eye  $h_e = 1.6$  m up is only  $\sqrt{2Rh_e/(1 - k)} \approx 4.8$  km away. Cars 30 km away are visible from Marfa only because the platform looks down into a basin and US-67 climbs the far side (terrain cross-sections along several bearings are shown in the Supporting Information<sup>8</sup>). Whether a given car is visible is then decided by the terrain in between, and the margins are small: the ray from a lamp at 30 km, refracted with  $k = 0.13$ , rises only  $kD^2/8R \approx 2.3$  m above the straight chord at its midpoint. That is comparable to the height of a headlamp (0.66 m), much larger than the lidar error, and comparable to the 2-m error of a 30-m elevation model. This estimate sets the requirements for everything that follows.

### B. Exact geometry and the critical refraction coefficient

We place the eye  $O$ , the lamp  $Q$ , and terrain samples  $P_i$  along the geodesic from  $O$  to  $Q$  (computed with Karney’s algorithm on the WGS84 ellipsoid<sup>13</sup>) in Earth-centered Cartesian coordinates. For each sample we record its distance along the chord,  $s_i = (\mathbf{P}_i - \mathbf{O}) \cdot \hat{\mathbf{u}}$ , and its height above the chord,  $y_i = (\mathbf{P}_i - \mathbf{O}) \cdot \hat{\mathbf{w}}$ , where  $\hat{\mathbf{u}}$  points from  $O$  to  $Q$  and  $\hat{\mathbf{w}}$  is perpendicular to it in the vertical plane. Earth’s curvature is then contained exactly in the  $y_i$ ; no  $d^2/2R$  approximation is needed.

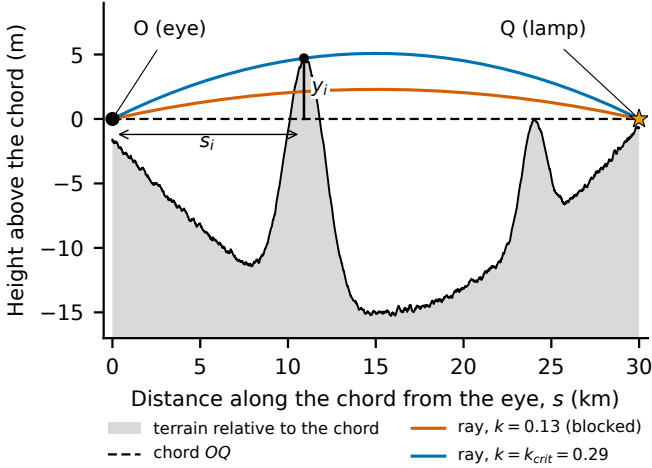


FIG. 2. Line-of-sight geometry in the frame of the straight chord  $OQ$  (illustrative profile, 30 km). The terrain is plotted as its height  $y$  above the chord, which includes Earth’s curvature. A refracted ray with coefficient  $k$  lies above the chord by  $ks(D-s)/2R$ . Here the ridge at  $s_i$ , which rises  $y_i$  above the chord, blocks the standard ray ( $k = 0.13$ ); the lamp becomes visible for  $k \geq k_{\text{crit}} = 0.29$ .

With a constant refraction coefficient the ray from  $O$  to  $Q$  is a circular arc of radius  $R/k$  (Appendix A), which lies above the chord by

$$\sigma_i(k) = \frac{k s_i(D - s_i)}{2R}, \quad (2)$$

where  $D = |OQ|$  and  $R$  is the radius of curvature of the ellipsoid’s normal section in the direction of the sight line (Euler’s formula). The lamp is visible if the arc clears every sample,  $y_i \leq \sigma_i(k)$  for all  $i$ . Because  $\sigma_i$  is linear in  $k$ , this is equivalent to

$$k \geq k_{\text{crit}} \equiv \max_i \frac{2R y_i}{s_i(D - s_i)}. \quad (3)$$

Figure 2 illustrates this. Equation (3) turns a visibility question into a single number per lamp position: a lamp with  $k_{\text{crit}} < 0$  is visible even with a straight ray, one with  $k_{\text{crit}} = 0.4$  appears only in air more strongly refracting than standard, and so on. We sample the terrain every 2 m within 3 km of either end and every 5 m in between, and skip the first 5 m at the eye and the last 3 m at the lamp (the observer’s own feet and the car’s body). Halving the step changes no result.

### C. How robust is “visible”?

A visible/hidden answer from Eq. (3) is only as good as its inputs, so we attach four robustness measures to each lamp position.

*Margins.* The lamp margin  $d_L$  is the smallest change in lamp height that flips visibility, and  $d_E$  is the same for

the eye. On US-67 no visible point has a lamp margin larger than 0.84 m.

*Terrain error.* We draw 200 realizations of correlated Gaussian terrain error along each sight line, with a standard deviation and correlation length for each elevation source: lidar 0.056 m (its accuracy report) with an assumed 20 m; 1/3 arc-second 0.56 m and 70 m; 1 arc-second 2.05 m and 125 m, the latter two measured against the lidar on 40 000 points. A 0.15 m error in eye height is added. The fraction of realizations in which the lamp is visible,  $P_{\text{vis}}$ , gives three classes: visible ( $P_{\text{vis}} \geq 0.95$  and still visible with 0.25 m of unresolved grass added to the ground), hidden ( $P_{\text{vis}} \leq 0.05$ ), and marginal (the rest). In Table I, “robust” uses the  $P_{\text{vis}}$  criterion alone. The Supporting Information gives details.<sup>8</sup>

*Obstructions.* From the lidar point cloud (29.4 million points around the grazing corridors) we mark a 1-m cell as obstructed when it has at least two returns and at least 25% of them more than 0.3 m above the ground, and raise it to the height of its highest return. Ground returns agree with the elevation model to a median of  $-0.8$  mm.

*Refraction.* Section IV.

The code reproduces an earlier, independent implementation (1075 of 1077 points agree) and passes 45 unit tests, 15 of which re-derive every equation of this paper independently of the model code; both checks are described in the Supporting Information.<sup>8</sup>

## IV. REFRACTION NEAR THE GROUND AT NIGHT

A single  $k$  is a poor description of rays that graze the ground at night. For a ray at small elevation, the local coefficient follows from the vertical temperature gradient,<sup>12</sup>

$$k = 503 \frac{P}{T^2} \left( 0.0343 + \frac{dT}{dz} \right), \quad (4)$$

with  $P$  in hPa,  $T$  in K, and  $dT/dz$  in K/m. At the site ( $P \approx 850$  hPa,  $T \approx 283$  K) a neutral atmosphere ( $dT/dz = -0.0098$  K/m) gives  $k = 0.13$ , an isothermal one  $k = 0.18$ , and the gradients of 1–2 K/m that Hirt *et al.* measured 1.8 m above grass after sunset give  $k$  of 5–11.<sup>12</sup> We therefore also trace rays through temperature profiles that follow the terrain. In the chord frame, with elevation angles below  $1^\circ$ , a ray obeys

$$y''(s) = -\frac{k(a(s))}{R}, \quad a = y - y_{\text{ground}}(s), \quad (5)$$

where  $a$  is the ray’s height above the local ground. For constant  $k$  the ray through  $Q$  is exactly Eq. (2), which we use as a test. We launch a fan of rays from the eye, discard those that hit the terrain, and find the launch angles that arrive at the lamp; each is an image. The profiles are brackets, not a climatology: an exponential stable boundary layer with 3, 6, or 10 K of inversion over 50, 30, or 20 m (the standard nocturnal form<sup>14</sup>; surface

gradients 0.06, 0.2, and 0.5 K/m), an isothermal layer, a neutral one, and an elevated 4-K inversion between 40 and 60 m. They do not reach the strongest gradients measured in the lowest 2 m, which the rays at issue here cross only near the two ends.

A useful bound limits the work. If the largest local coefficient anywhere along a path is  $k_{\max}$ , then  $y'' \geq -k_{\max}/R$  with fixed end points, so every ray lies below the constant- $k_{\max}$  arc. A lamp with  $k_{\text{crit}} > k_{\max}$  is therefore hidden in that atmosphere, and need not be traced. In our profiles  $k_{\max} = 2.82$ , which leaves 212 of the 1548 US-67 points (Marfa to Presidio, 60-m spacing) to trace.

## V. PHOTOMETRY AND DETECTION

### A. Where the platform falls in the beam

Lamps are strongly directional, so brightness depends on where the observer sits in the lamp's own coordinates. For a car with heading  $\psi$  we define the horizontal angle of the observer off the heading, positive to the driver's right,  $h = \beta - \psi$  (with  $\beta$  the bearing from car to observer), and the vertical angle above the lamp axis,

$$v = \epsilon_Q - \arctan g - \delta_{\text{aim}}, \quad \epsilon_Q = -e_O - \frac{D}{R} + \frac{kD}{2R}. \quad (6)$$

Here  $e_O$  is the elevation of the chord at the eye, so  $\epsilon_Q$  is the elevation of the ray at the lamp above the lamp's local horizontal (the verticals at the eye and the lamp differ by the angle  $D/R$ );  $g$  is the road grade along the heading, which pitches the car; and  $\delta_{\text{aim}}$  is the vertical aim error of the lamp, positive when it is aimed high (zero in the central case,  $\pm 0.5^\circ$  in the bands). We take  $g$  from the lidar  $\pm 25$  m along the road. On US-67 the median grade is 1.6% and the 95th percentile 4.7%, so pitch commonly moves the observer in the beam by  $1^\circ$ – $3^\circ$ , as much as the geometry itself.

### B. Lamp intensities

For headlamps we use the market-weighted U.S. beam patterns of the University of Michigan Transportation Research Institute: 25th, 50th, and 75th percentiles of luminous intensity on a grid from  $45^\circ$  left to  $45^\circ$  right and  $5^\circ$  down to  $7^\circ$  up, for low beams of the 20 best-selling vehicles of model year 2004 and for high beams of the same era.<sup>15,16</sup> The median low beam has 8830 cd straight ahead, but only 55–100 cd at  $30^\circ$  and a sharp upper cut-off above the horizontal; the median high beam has 37 000 cd straight ahead. We interpolate in  $\log I$ . No comparable survey exists for tail and stop lamps, so we bracket them by the minimum and maximum of Federal Motor Vehicle Safety Standard No. 108: a tail lamp must give at least 2.0 cd on axis (0.8 cd at  $10^\circ$ , 0.3 cd at  $20^\circ$ ) and at most 18 cd; a stop lamp 80–300 cd; the high-mounted stop lamp 25–160 cd.<sup>17</sup>

### C. Magnitude and threshold

At these ranges the two lamps of a car (1–1.5 m apart,  $7''$ – $10''$  at 30 km) are an unresolved point, and the illuminance at the eye is

$$E = \frac{\sum_j I_j(h, v) \mathcal{T}}{D^2}, \quad \mathcal{T} = \exp\left(-\frac{D \ln 20}{V}\right), \quad (7)$$

where  $V$  is the meteorological optical range (the distance at which transmission falls to 5%,  $V = \ln 20/b$  for extinction coefficient  $b$ ). The National Park Service reports a standard visual range at nearby Big Bend National Park of about 90 miles on average, below 55 miles on high-pollution days, and about 165 miles without pollution.<sup>18</sup> Standard visual range is defined by a 2% contrast threshold,  $3.912/b$ , so  $V$  is 0.766 times it: 111, 68, and 203 km, which we use as reference, least, and most favorable values. The apparent magnitude of a point source of illuminance  $E$  (in lux) is<sup>19</sup>

$$m = -13.99 - 2.5 \log_{10} E. \quad (8)$$

For example, a car 30 km away that sends 300 cd per lamp toward the platform, with  $V = 111$  km, gives  $E = 3.0 \times 10^{-7}$  lx and  $m = +2.3$ , about as bright as Polaris.

Whether the eye sees it depends on the background. Crumey's fit to laboratory contrast thresholds gives the naked-eye limiting magnitude of a point source on a background of surface brightness  $\mu$  (in mag arcsec $^{-2}$ ) as<sup>20</sup>

$$m_{\text{lim}} = 0.3834 \mu - 1.4400 - 2.5 \log_{10} F, \quad (9)$$

valid for  $20 < \mu < 22$ , where  $F$  is a field factor that describes the observer and conditions (typically 1.4–2.4). The sky brightness at Marfa's horizon has not been measured, so we span  $\mu = 20.5$ – $22$  and  $F = 1.4$ – $4$ , which gives  $m_{\text{lim}} = 4.9$ – $6.6$  (reference 5.9 for  $\mu = 21$ ,  $F = 2$ ).

## VI. RESULTS

### A. What is in view

Table I gives the length of US-67 in view for a headlamp 0.66 m high at  $k = 0.13$ , from Marfa to Presidio, for each terrain model. The three elevation models agree point by point for 98.6–99.7% of the road, but only the lidar is accurate enough to decide visibility with confidence: with the 1-arc-second model almost all of the “visible” road is uncertain in the Monte Carlo, with the lidar almost all of it is robust. Measured vegetation and structures remove about 0.5 km. Of the 64.5 km between Shafter and Marfa, 9.5 km is in the visible class, 0.5 km marginal, and 54.5 km hidden. At 30-m sampling the visible road forms 29 separate stretches (five of them single samples; at the 60-m sampling used below, 18), 18–40 km from the platform, at bearings  $229^\circ$ – $238^\circ$  (plus one 60-m point at  $251.9^\circ$ ), at apparent elevations

TABLE I. Length of US-67 (Marfa to Presidio, 3095 points every 30 m) in view of the platform for a lamp 0.66 m high at  $k = 0.13$ . “Visible”: deterministic,  $k_{\text{crit}} \leq k$ ; “robust”:  $P_{\text{vis}} \geq 0.95$ ; “uncertain”:  $0.05 < P_{\text{vis}} < 0.95$ .

| Terrain model         | Visible<br>(km) | Robust<br>(km) | Uncertain<br>(km) |
|-----------------------|-----------------|----------------|-------------------|
| 1 arc-second DEM      | 10.26           | 0.00           | 9.81              |
| 1/3 arc-second DEM    | 10.26           | 0.21           | 10.71             |
| 1-m lidar, bare earth | 10.44           | 10.26          | 0.21              |
| Lidar + obstructions  | 9.93            | 9.72           | 0.27              |
| + 0.25 m grass        | 9.36            | –              | –                 |
| + 1 m uniform clutter | 2.04            | –              | –                 |

between  $-0.40^\circ$  and  $+0.06^\circ$ . Every one of them lies below the skyline, by  $0.04^\circ$ – $0.84^\circ$  (median  $0.26^\circ$ ), so the lights are seen against the dark mountains rather than the sky. The limiting terrain point lies a median 3.0 km from the car.

Lamp height matters more than refraction: at  $k = 0.13$  a headlamp 0.56 m high sees 9.72 km of road, one at 1.37 m 10.83 km, and a truck’s clearance lamps at 4 m 13.77 km (the full table is in the Supporting Information<sup>8</sup>).

## B. The view from the platform

Figure 3 shows every catalogued ordinary source as the observer sees it. Panel (a) spans  $155^\circ$ – $300^\circ$  of bearing, which includes the usual  $120^\circ$  fan, with the vertical scale exaggerated so that the structure below the skyline is visible. Panel (b) shows the US-67 sector at true scale, as the eye sees it: the possible positions of cars on the highway form a broken chain against the dark terrain, typically a quarter of a degree (and at most  $0.84^\circ$ ) below the skyline, and nowhere rise above it. Each gap in the chain is a stretch of road hidden by a fold of the basin. Seen at this scale, a single car appearing in one window and then another could easily be described as a light that moves, vanishes, and reappears elsewhere.

## C. Refraction

Refraction matters more for where a light appears than for whether it appears. Raising  $k$  from 0.13 to 1 brings about 1.6 km more of US-67 into view, but a change  $\Delta k$  lifts a light at distance  $d$  by  $d\Delta k/2R$ ,  $0.13^\circ$  for  $\Delta k = 1$  at 30 km. In the ray tracing, the strongest of our surface inversions adds 1.5 km of visible road and lifts lights by at most  $7.5'$ . The elevated inversion produces three to five images of one lamp at five road points, a superior mirage, but they are only 1–8 arcseconds apart, far below the eye’s resolution; they would be seen as one light, possibly twinkling. Surface layers whose gradient decreases monotonically with height gave single images

TABLE II. Predicted magnitude of one car on US-67 (Shafter–Marfa) where its lamps are in view, and the road length over which it is brighter than  $m_{\text{lim}}$ : reference ( $V = 111$  km,  $\mu = 21$ ,  $F = 2$ ,  $m_{\text{lim}} = 5.9$ ) and full bracket (least favorable  $V = 68$  km,  $\mu = 20.5$ ,  $F = 4$ ; most favorable  $V = 203$  km,  $\mu = 22$ ,  $F = 1.4$ ). Road sampled every 60 m; “in view” means  $P_{\text{vis}} \geq 0.5$  for the lamp’s height.

| Lamps                              | $m$ , median<br>(range) | Detectable<br>(km) | Bracket<br>(km) |
|------------------------------------|-------------------------|--------------------|-----------------|
| <i>Northbound, 10.0 km in view</i> |                         |                    |                 |
| Low beam, 25th pct.                | 2.9 (−1.4 to 6.8)       | 9.4                | 8.7–10.0        |
| Low beam, median                   | 2.6 (−1.7 to 6.0)       | 9.6                | 9.0–10.0        |
| High beam, median                  | 0.4 (−2.4 to 5.4)       | 10.0               | 9.3–10.0        |
| <i>Southbound, 10.1 km in view</i> |                         |                    |                 |
| Tail, FMVSS min.                   | 9.1 (7.9 to 10.6)       | 0                  | 0               |
| Tail, FMVSS max.                   | 5.6 (4.7 to 6.3)        | 6.7                | 0–10.1          |
| Braking, FMVSS min.                | 4.6 (3.7 to 6.1)        | 9.4                | 4.4–10.1        |

only. The results for every profile are tabulated in the Supporting Information.<sup>8</sup>

## D. Brightness

US-67 runs almost straight at the platform. A northbound car (toward Marfa) has the platform  $3^\circ$ – $33^\circ$  to the driver’s right (5th–95th percentile; median  $14^\circ$ ), and  $0.6^\circ$  below to  $2.5^\circ$  above the lamp axis, near the top of the low beam [Fig. 4(a,b)]. Table II and Fig. 4(c) give the result. A northbound car is a naked-eye light over at least 87% of the road in view for every lamp percentile, visual range, and threshold we tried, and over 93% at the reference values: typically magnitude  $+2.6$  on low beam and  $+0.4$  on high beam. On the straight at  $233.7^\circ$  (31–33 km) it reaches  $-1.7$  on low beam and  $-2.4$  on high beam, brighter than Sirius ( $-1.46$ ). A southbound car shows its tail lamps  $3^\circ$ – $33^\circ$  off its rear axis (5th–95th percentile). At the regulatory minimum they are invisible ( $m \approx 9$ ); at the maximum they are at the threshold ( $m \approx 5.6$ ); braking brings them to  $m \approx 4.6$  even at the minimum, visible over most of the road. Apart from the fixed red light of the tower at  $230.3^\circ$  (Table III), a red light on these bearings is therefore most likely a braking car.

## E. One car in time

Figure 5 follows a northbound car at 30 m/s. It takes 13.6 minutes to cross the region in view and is visible for 334 s of it, in 18 separate windows: a median 510 m of road (17 s), from 60 m (2 s, our sampling limit) to 2.1 km (70 s). While visible it drifts across the view at about  $0.9^\circ$  per minute, never more than  $0.84^\circ$  below the skyline. Within a window its brightness changes by a median of 1.1 magnitudes (up to 4.9), and between successive 2-s samples by up to 0.8 magnitudes in one case in ten, with

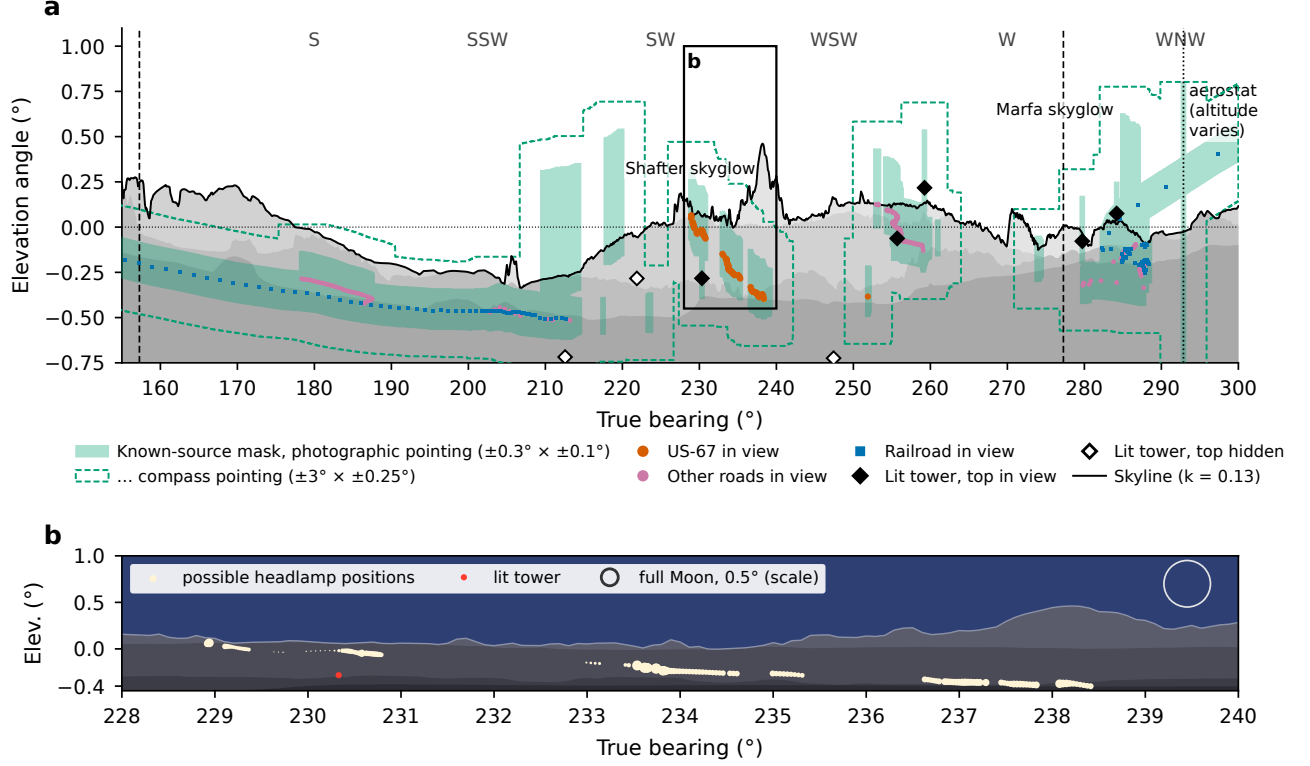


FIG. 3. (a) Catalogued ordinary light sources as seen from the viewing area ( $k = 0.13$ ), with the known-source mask: the set of directions in which some catalogued source can appear for any  $k$  from 0 to 1, widened by the observer's pointing error (filled: photographic pointing,  $\pm 0.3^\circ \times \pm 0.1^\circ$ ; dashed: compass pointing,  $\pm 3^\circ \times \pm 0.25^\circ$ ). Bearings  $155^\circ$ – $300^\circ$  true, vertical scale exaggerated 22 $\times$ . Grey shades are ridges within 10, 25, and 45 km and beyond; dashed vertical lines mark the usual  $120^\circ$  viewing fan, and the dotted line the bearing of the tethered aerostat, whose altitude varies. The box is the region shown in (b). (b) The US-67 sector,  $228^\circ$ – $240^\circ$  by  $-0.45^\circ$  to  $+1.0^\circ$ , at true scale (no vertical exaggeration) and rendered as at night. Each white dot is a position where a northbound car's headlamps can appear, with area increasing with the predicted brightness (low beam, fleet median,  $V = 111$  km); at any moment only a few of these positions are occupied (Sec. VIE). The red dot is the obstruction light of the 89-m tower at  $230.3^\circ$ , 20 km away; the circle shows the size of the full Moon. An animation of one car crossing this scene is on the paper's web page.

no change in the car: grade changes pitch the beam by degrees across the top of the low beam, where intensity falls by a factor of about 2 per degree, and curves swing the heading.

One ordinary car therefore produces a light that appears, drifts, brightens and dims, vanishes, and reappears a little further on, 18 times. (The 18 windows are the 29 stretches of Sec. VIA sampled every 60 m instead of 30 m.) For a one-way flow  $q$  with random arrivals, the number of cars in view is Poisson distributed with mean

$$N = \frac{qL}{u}, \quad (10)$$

where  $L$  is the length of road in view and  $u$  the speed. With  $L = 10.0$  km and  $u = 30$  m/s,  $N = 0.93$  per 10 vehicles per hour in each direction. At the daily-mean flow implied by the AADT (23–29 vehicles per hour each way), two or three northbound headlights would be in view at any moment; at night the flow is lower by a factor we have not measured.

## F. The other catalogued sources

Table III collects the observable signatures of every catalogued source in the southwestern view, the information an observer needs to recognize each one. Besides US-67, the catalogue contains other roads, two railroads, lit towers, the glow of towns, and a tethered aerostat. Ranch-to-Market Road 2810 is in view for 13 km of its 52 km, at  $253^\circ$ – $259^\circ$  and 32–53 km, but carries fewer than 60 vehicles a day. One county road, Nopal Road, which runs south across the flat from US-90, is in view for about 5 km at  $178^\circ$ – $213^\circ$  and 7–13 km; it has no traffic count, and we assume 60 vehicles a day. US-90 and the US-67/90 concurrency beside the platform are in view at  $280^\circ$ – $288^\circ$ , at 0.3–26 km. The Texas Pacific railroad is in view for 6.7 km within the usual fan ( $159^\circ$ – $213^\circ$ , 3.8–8 km; 12.5 km in all) for a locomotive headlight 4 m high, and the crossing inventory reports at most one through train per night on it;<sup>21</sup> the Union Pacific line, with up to six night trains, is in view at  $282^\circ$ – $297^\circ$ , partly above the

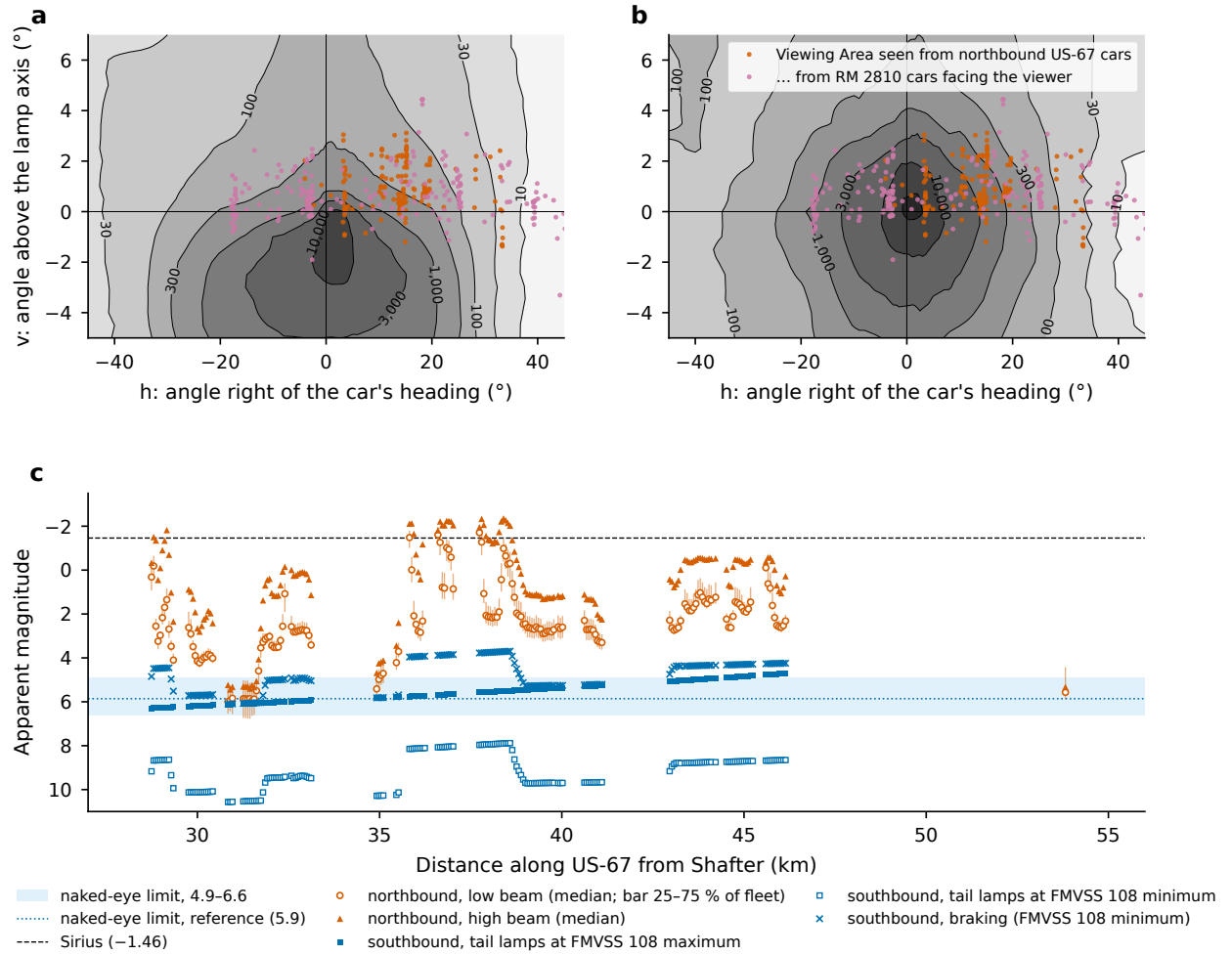


FIG. 4. (a) Median U.S. low-beam and (b) high-beam pattern of one lamp (UMTRI market-weighted data; contours in candela), with the platform's position in the beam of northbound cars on US-67 (orange) and of cars on RM 2810 facing the platform (pink). (c) Predicted magnitude of one car (two lamps) on the stretches of US-67 in view, versus distance along the road from Shafter; only road in view is plotted ( $V = 111$  km,  $k = 0.13$ ): northbound headlamps (circles: low beam, bars 25th–75th percentile of the fleet; triangles: high beam) and southbound tail lamps (filled squares: regulatory maximum; open: minimum; crosses: braking at the minimum). Band: naked-eye limit, Eq. (9).

skyline. Of the 65 registered towers in the area, 29 are lit; seven of these show their top light, five within  $155^\circ$ – $300^\circ$  and three within the usual fan, with red obstruction lights, steady or flashing at a fixed rate.<sup>22</sup> The glow of Shafter, Presidio, Ojinaga, and Marfa sits on the skyline at fixed bearings.

Combining these sources with traffic rates gives a map of how often an ordinary light is expected in each direction. Within the usual fan and within 5 mrad ( $0.29^\circ$ ) below the skyline, 77% of the view has an expected rate of catalogued transient lights below 0.01 per hour and no fixed light. This estimate assumes that 2% of the daily traffic passes in a night hour, that a quarter of drivers use high beams, and that a light's position is known to  $\pm 0.3^\circ$  in azimuth and  $\pm 0.1^\circ$  in elevation. With compass bearings ( $\pm 3^\circ$  by  $\pm 0.25^\circ$ ) the quiet fraction falls to 36%, and the known-source mask of Fig. 3(a) grows from 33%

to 74% of the band. The precision of a report therefore decides how much of the view it can test.

### G. A check against photographs

The geometry can be checked against photographs taken from the platform. On 21 November 2018, 16–30 minutes after sunset, the author photographed the US-67 sector with a 210-mm lens on an APS-C camera (nominally 936 pixels per degree).<sup>23</sup> We registered each frame by fitting a four-parameter camera model (azimuth, elevation, roll, scale) to the modeled skyline, with residuals of  $0.005$ – $0.017^\circ$ , and then located point lights in the short exposures with a detection that does not use the road model [Fig. 6(a)]. Grouping detections in frames seconds apart gives eight independent sightings. Their

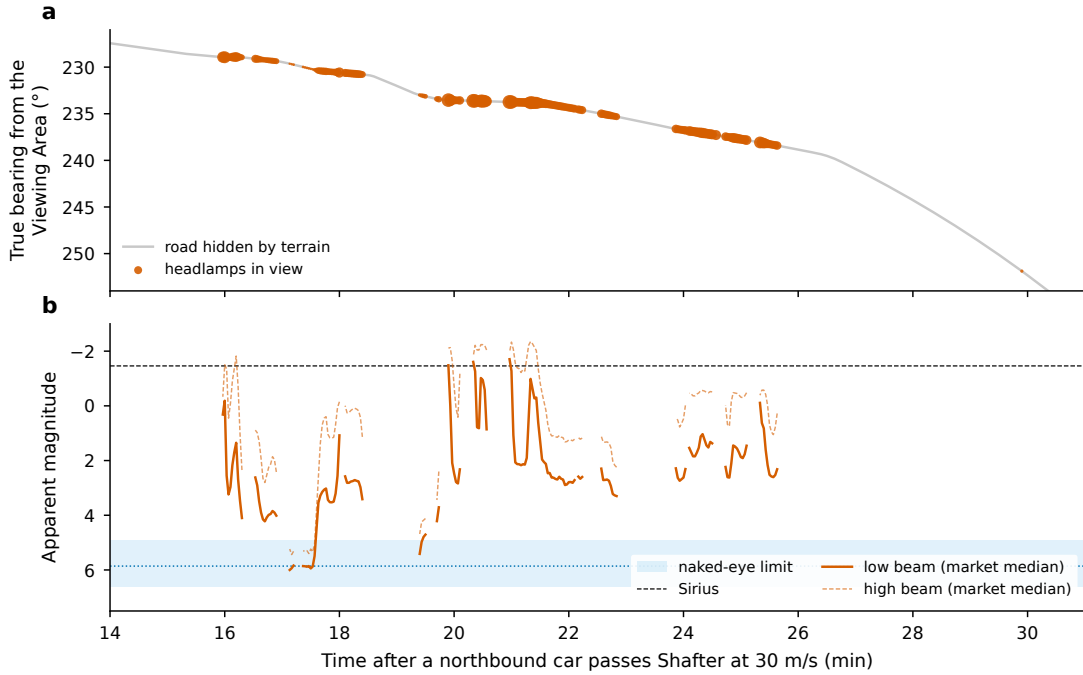


FIG. 5. A northbound car on US-67 at 30 m/s, seen from the platform ( $V = 111$  km,  $k = 0.13$ ). It appears in 18 windows, for a median of 17 s each and 334 s in all, during the 13.6 min it takes to cross the region in view. (a) Bearing versus time; orange where the headlamps are in view (dot area increases with brightness), grey where terrain hides the road. (b) Magnitude on low beam (solid) and high beam (dashed), fleet median,  $V = 111$  km.

median angular distance from the modeled visible road is  $0.0085^\circ$  (8 pixels, about 4 m at 26 km), whereas random positions in the same band of the frames have a median distance of  $0.28^\circ$ ; the probability that at least half of eight random positions fall that close is  $3 \times 10^{-6}$  [Fig. 6(c)]. About half of the road that falls in these frames is modeled as hidden, yet all eight sightings lie on visible stretches; if the visible/hidden split had no predictive power, the chance of this would be the product of each frame's visible fraction, about 0.005. One light seen in two frames 6 s apart moved  $179 \pm 30$  m along the road, 30 m/s (17–48 m/s at  $2\sigma$ ). A 45-s exposure records the traffic as a streak that follows the modeled road [Fig. 6(b)].

This is a test of internal consistency, not of absolute pointing: because the frames are registered to the modeled skyline, an error common to skyline and road would not show. It also does not test brightness (the frames are not photometrically calibrated) or refraction (the elevation zero point is degenerate with  $k$ ).

## VII. USING THE MODEL TO SCREEN OBSERVATIONS

The results above turn into a sequence of tests that removes ordinary lights from a set of observations, in the order of how much each test removes.

1. *Position.* Is the light inside the known-source mask for the precision with which its position was measured?

A light seen above the skyline is compared with aircraft tracks, with star, planet, and satellite ephemerides, and with the few fixed sources that reach above the skyline (two towers, the aerostat, and parts of the Union Pacific line); one below it is compared with Table III.

2. *Color and brightness.* Does it match the source class at that position: white and of magnitude about  $-2$  to 6 on the US-67 bearings for a car approaching Marfa; red and near the limit of vision for one driving away; red and fixed for a tower?

3. *Motion.* Does it move along the locus of a road at a rate a vehicle there could produce ( $\approx 0.9^\circ$  per minute on US-67, toward larger bearing for northbound and smaller bearing for southbound traffic), or stay fixed like a tower?

4. *Timing.* Does it last for seconds to about a minute and then reappear a little further along the road, as a car passing through the windows of Fig. 5 does? Is the number of such lights consistent with the traffic, Eq. (10)?

A light that matches no catalogued source on any one of these tests—one outside the mask, or one inside it whose color, brightness, motion, or timing fits no source there—is a candidate worth careful measurement. Examples are a light that holds its position for many minutes where there is no tower, moves across a road's locus or faster than a vehicle on it could, rises above the skyline with no aircraft to account for it, or appears in a direction where no catalogued source can be seen at all. The model does not say what such a light is; it says that it is not one of the ordinary sources in Table III.

TABLE III. Observable signatures of the catalogued ordinary light sources in the southwestern view from the viewing area (model,  $k = 0.13$ ,  $V = 111$  km). Bearings are true. Magnitudes are for one vehicle; the naked-eye limit is  $m_{\text{lim}} \approx 4.9$ –6.6.

| Source                               | Bearing                                | Elevation                                                                                             | Color, brightness                                                                | Behavior                                                                          | Frequency                                          |
|--------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------|
| US-67 northbound (headlamps)         | 229°–238° (and 252°)                   | 0.04°–0.84° below the skyline                                                                         | white; $m$ from $-1.7$ to $6$ , median $2.6$ (low beam)                          | 18 windows of 2–70 s; drifts $\approx 0.9^\circ/\text{min}$ toward larger bearing | $\approx 1$ in view per 10 vehicles/h              |
| US-67 southbound (tail lamps)        | same                                   | same                                                                                                  | red; $m$ 4.7–6.3 at FMVSS maximum, invisible at minimum; braking 3.7–6.1         | same windows in reverse order; drifts toward smaller bearing                      | same                                               |
| RM 2810 (toward Marfa)               | 253°–259°                              | 0–0.27° below the skyline                                                                             | white; $m$ median 4.0                                                            | short windows                                                                     | $< 1$ vehicle/h                                    |
| Nopal Road (county road)             | 178°–213°                              | 0.14°–0.28° below the skyline                                                                         | white northbound ( $m$ $-1.3$ to $3.5$ , low beam), red southbound; 7–13 km away | crosses the view in either direction                                              | no count; assumed $\approx 0.6$ vehicle/h each way |
| US-90 and US-67/90                   | 280°–288°                              | near horizontal                                                                                       | white or red; very bright                                                        | fast; 0.3–26 km away                                                              | busiest roads                                      |
| Texas Pacific railroad               | 159°–213°                              | 0.2°–0.5° below horizontal                                                                            | white headlight and ditch lights                                                 | slow; 3.8–8 km away                                                               | $\leq 1$ train per night                           |
| Union Pacific railroad               | 282°–297°                              | partly above the skyline                                                                              | white headlight and ditch lights                                                 | 0.3–27 km away                                                                    | $\leq 6$ trains per night                          |
| Lit towers, top in view              | 230.3°, 255.7°, 259.2°, 279.7°, 284.2° | fixed: $-0.34^\circ$ , $-0.19^\circ$ , $+0.08^\circ$ , $-0.04^\circ$ , $+0.06^\circ$ from the skyline | red obstruction light                                                            | steady or fixed flash rate                                                        | permanent                                          |
| Town skyglow                         | 211°–219°, 286°                        | on the skyline                                                                                        | diffuse glow                                                                     | fixed                                                                             | permanent                                          |
| Tethered aerostat                    | 292.9°                                 | variable                                                                                              | lights when aloft                                                                | fixed bearing                                                                     | intermittent                                       |
| Aircraft, satellites, stars, planets | any                                    | mostly above the skyline                                                                              | various                                                                          | predictable from flight tracking and ephemerides                                  | continual                                          |

The same model is available as a website, <https://zacharyslate.github.io/marfa-lights-investigation/>, which applies the position test and lists the candidate sources, with their expected brightness and rate, for any direction an observer enters. It includes an interactive map of the view with every catalogued source, a phone-camera overlay that draws the skyline and the predicted positions of roads and towers on the live image and records calibrated bearings, and a sighting form that checks a report against every known source. Photographs like those of Fig. 6 can be registered to the modeled skyline with the published code.

## VIII. DISCUSSION

*What the model accounts for.* Several features often reported for the lights follow from ordinary physics once the geometry is known. Lights on the US-67 bearings sit just below the skyline because the only road in view lies on the far slope of the basin, seen against the mountains. They appear and vanish without moving much because terrain cuts the road into short windows. They can be brighter than any star where the road points at the platform and a car pitches its beam upward. They brighten and fade within a single appearance because grade and curvature move the observer across the low-beam pattern. Car lights on these bearings are hard to recognize as such because the eye has no distance cue for a point

source, and a light 30 km off at magnitude  $+2$  looks like a star. Refraction adds only minutes of arc of lift and, in our profiles, no resolvable splitting. Recognizing these lights for what they are is what allows any others to stand out.

*What it does not account for.* A model of where ordinary lamps can appear does not show that any particular sighting was one of them, and it cannot describe sources that are not in its catalogue: private ranch roads and ranch lights, off-road vehicles, and anything not yet known. The unexplained part of the view is therefore unexplained only with respect to the catalogue; extending the catalogue, for example with ranch lighting mapped from satellite night imagery, would shrink it further. Conversely, lights that remain outside the mask after careful measurement are exactly the observations that deserve attention.

*Limitations.* The largest are in the photometry. The beam survey describes lamps of about 2004; modern LED headlamps have sharper cut-offs and different wide-angle light, which the fleet percentiles do not cover. Signal lamps are bracketed by regulation, not measured. Pitch is approximated by the road grade, and the horizon sky brightness and the observer’s field factor are not known for this site. The headlamp conclusion is insensitive to these; the tail-lamp result is not. In the geometry, grass lower than about 0.3 m is below what the lidar resolves (allowing for 0.25 m of it removes 0.6 km of visible road), and the refraction profiles are brackets, not measure-

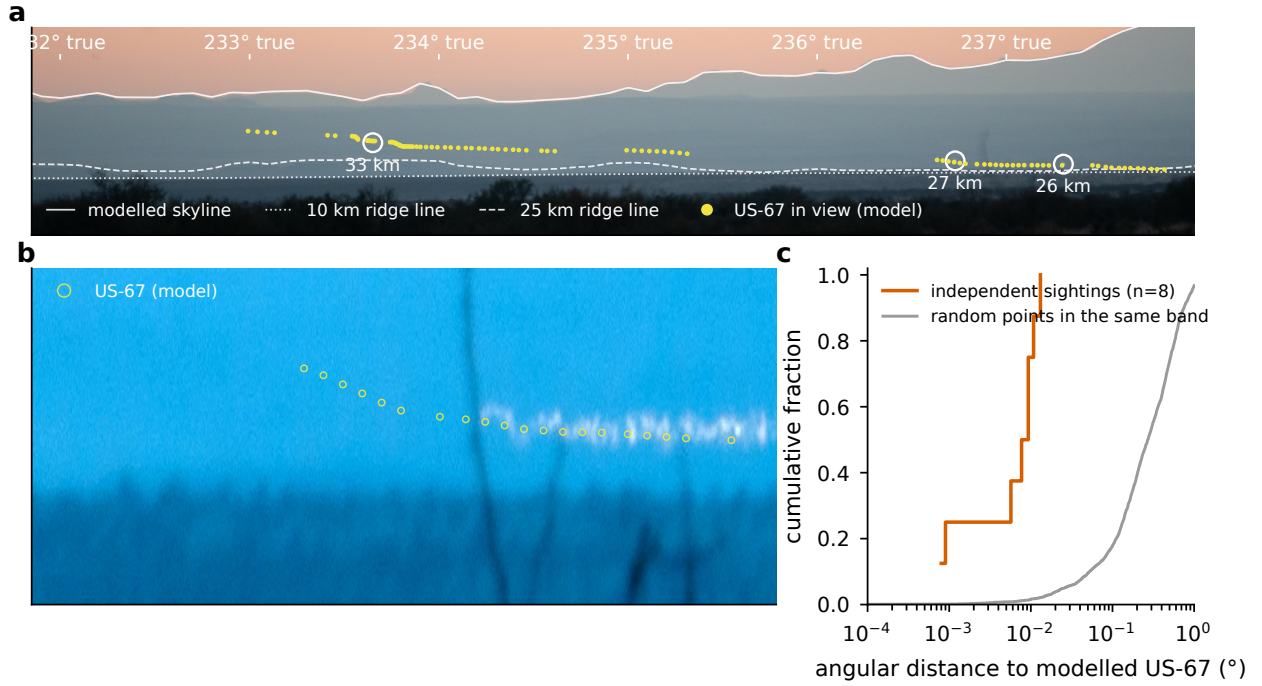


FIG. 6. Check of the geometry against photographs from the platform, 21 November 2018. (a) Detected lights (circles, with distances) and the modeled US-67 in view (dots), with the modeled skyline and ridge lines. (b) A 45-s exposure; the streak of traffic, found without reference to the model, lies a median  $0.005^\circ$  from the modeled road (circles). (c) Cumulative distribution of the distance from detected lights to the modeled road, and of random positions in the same band.

ments.

*For students.* Each piece of the model can be done by hand or in a short program: the curvature drop and horizon distance [Eq. (1)]; the critical coefficient for a single ridge [Eq. (3)]; the magnitude of a car of known intensity [Eqs. (7) and (8)]; the occupancy of a road window [Eq. (10)]. A class with access to a dark roadside viewpoint can repeat the whole analysis for a local “ghost light” with free national elevation data.

## IX. CONCLUSIONS

We have built a quantitative model of the ordinary lights seen from the Marfa lights viewing area, intended as a tool for separating them from anything else. Exact geometry on 1-m lidar terrain shows that about 10 km of US-67, in 29 short stretches 18–40 km away, is in view of the platform, always just below the skyline. A single number, the critical refraction coefficient, decides visibility for each lamp position; because rays graze the ground within about a meter, only lidar terrain with a terrain-error Monte Carlo makes that decision robust, and nocturnal inversions of up to  $0.5 \text{ K/m}$  at the surface change it little. Market-weighted beam patterns and a contrast-threshold model show that a northbound car there is a naked-eye light over nearly all of the road in view, typically of second or third magnitude and sometimes brighter than Sirius, while tail lamps are near the limit

of vision. A single car should appear some 18 times in a quarter of an hour, brightening and fading with the road’s grade and curves. Together with the railroad, the lit towers, and the glow of towns, these predictions give a table of observable signatures and a map of where ordinary lights can and cannot appear. Used as a screen, the model lets observers remove false positives and concentrate on the lights that remain unexplained.

## AUTHOR DECLARATIONS

*Conflict of interest.* The author has no conflicts to disclose.

*Data availability.* This paper and its Supporting Information are published at <https://zacharyslate.github.io/marfa-lights-investigation/paper.html>. The model is available as an interactive website at <https://zacharyslate.github.io/marfa-lights-investigation/>. All code, derived data, and figures, the public inputs, and a manifest of the terrain and geoid files (URLs and SHA-256 checksums) are available at <https://github.com/zacharyslate/marfa-lights-investigation>; each released version of the paper is tagged in the repository and archived on Zenodo with a DOI. The photographs are available from the author on request.

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tribution 4.0 International licence (CC BY 4.0, <https://creativecommons.org/licenses/by/4.0/>) and the code under the MIT licence. The photographs in Fig. 6 are excluded; all rights to them are reserved.

*Use of AI tools.* We disclose that an AI system (Claude, Anthropic; configured model identifier claude-opus-5-5, 2026) was used to write and test the analysis code, to extract tabulated values from source documents (the UMTRI and FMVSS photometric tables, checked cell by cell against the printed tables), and to draft text. It was used because the project was carried out by a single author. All code is public and covered by unit tests, and every number in this paper is produced by that code.

## Appendix A: The refracted ray and the critical coefficient

In a horizontally stratified atmosphere with refractive index  $n(z)$ , a ray at elevation angle  $e$  bends toward the denser air with curvature  $\kappa = -(1/n)(dn/dz) \cos e$ . Writing  $\kappa = k/R$  defines the refraction coefficient; with  $n - 1 \propto P/T$  and hydrostatic pressure this gives Eq. (4). In the chord frame and for small angles,  $y'' = -k/R$ . With constant  $k$  and  $y(0) = 0$ , the ray is  $y = \theta s - ks^2/2R$ ; requiring  $y(D) = 0$  gives  $\theta = kD/2R$  and  $y = ks(D - s)/2R$ , Eq. (2). The relative error of the parabolic approximation is of order  $(D/R)^2 \approx 10^{-5}$ . The apparent elevation of the lamp is the chord elevation plus  $\theta$ , and a change  $\Delta k$  raises it by  $D\Delta k/2R$ .

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